

scheduled-work-engine — DESIGN

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1. Goal & scope

Backing infrastructure that gives the REMINDER / BACKGROUND action-prefix (§11.4.140) real teeth: a **decoupled, project-agnostic** (§11.4.28) service that tracks scheduled/background work items so an autonomous agent can, when a REMINDER fires, query for work whose outcome is **not yet confirmed** and re-verify it before reporting done (§11.4.6 no-guessing, §11.4.108 verify-before-report). No ATMOSphere/consumer-specific data is hardcoded; all configuration is via env/flags (§11.4.28).

2. Architecture

cmd/scheduled-work	subcommands: serve mcp-stdio
acp-stdio version	
└ internal/server	gin.Engine (REST + POST /mcp) +
brotli middleware + TLS	
└ internal/mcp	MCP JSON-RPC 2.0 dispatcher (shared
by HTTP + stdio)	
└ internal/acp	newline-delimited JSON-RPC stdio
loop (MCP-stdio + ACP-equiv)	
└ internal/store	JSON-file store (mutex + atomic
write), the data model	

One gin.Engine implements http.Handler, so it is served **identically** over HTTP/1.1, HTTP/2 (TCP TLS) and HTTP/3 (QUIC UDP). One MCP dispatcher backs the HTTP /mcp endpoint AND the stdio transports — a single tool implementation, three transports.

Current libraries (researched — §11.4.8/§11.4.99, verified 2026-07-02)

- github.com/quic-go/quic-go **v0.60.0** — `http3.Server{Handler, TLSConfig}` (`ListenAndServe / Serve(net.PacketConn)`) for the server; `http3.Transport` (an `http.RoundTripper`) for the client. `gin` does not serve HTTP/3 natively, so the `gin.Engine` is wrapped as the `http.Handler` behind `http3.Server` (confirmed against the vendored source, not assumed).
- github.com/andybalholm/brotli **v1.2.2** — `brotli.NewWriterLevel`; note it has no `WriteString` (the `gin.ResponseWriter WriteString` is implemented via `Write([]byte(s))`).
- github.com/gin-gonic/gin (latest) — router.
- github.com/google/uuid — item ids.

Graceful fallback: `serve` starts a TCP TLS `http.Server` (h1/h2) AND the UDP `http3.Server`; every h1/h2 response advertises the h3 alt-svc via `SetQUICHeaders`, so clients upgrade to HTTP/3 when able. `SWQ_H3=0` runs h1/h2 only.

3. Store choice — JSON file (justification, §11.4.6)

A reminder / scheduled-work queue is a **low-write-volume** workload. Options: - **SQLite (mattn/go-sqlite3)** — needs CGO, complicating static/cross builds. - **BoltDB** — an extra module dependency for a tiny dataset. - **JSON file (chosen)** — zero CGO, zero extra module, trivially portable across every supported platform (§11.4.81), human-inspectable for debugging.

Durability: a `sync.RWMutex` serialises all access; writes go through `write-temp-fsync-rename` (atomic replace), so a crash never leaves a torn file. Open on a corrupt file returns an error rather than silently dropping data (§11.4.6). The `TestConcurrentCreate` test drives 50 concurrent creates through the mutex with zero lost writes; `go test -race` is clean.

4. ACP surface — honesty note (§11.4.6 / §11.4.99)

“ACP” (Agent Client Protocol, agentclientprotocol.com) is an emerging JSON-RPC-2.0-over-stdio protocol whose full session / prompt-turn model targets interactive coding **agents**, not a stateless work-queue service. Implementing the complete ACP agent lifecycle here would invent behaviour that does not fit this service. Instead `internal/acp` exposes the **same** newline-delimited JSON-RPC-2.0-over-stdio transport ACP uses, dispatching to the identical tool set as the MCP surface. It is therefore an **ACP-EQUIVALENT** adapter, clearly labelled in source — not a claim of full ACP-agent conformance. This is stated as a fact, not hidden behind a green test.

5. Anti-bluff test coverage (§11.4 / §11.4.27 / §1.1)

- **Unit** (`internal/store`, `internal/mcp`) — model, lifecycle, derived overdue, needs-verification, persistence round-trip, invalid-input rejection, concurrency; MCP initialize / tools-list / tools-call / error paths.
- **Integration** (`test/`, no mocks — §11.4.27) — boots the REAL `http3.Server` on an ephemeral UDP port and round-trips `create` → `list` → `mark-done` → `verify` over **HTTP/3** (asserts `resp.Proto == "HTTP/3.0"`) with **brrotli** (asserts `Content-Encoding: br` and decodes it); plus the MCP-over-HTTP/3 endpoint.
- **§1.1 paired mutation** (`test/meta_mutation.sh`) — breaks the mark-done handler at source, asserts the suite goes RED, restores, asserts GREEN. Proves the suite is not a bluff gate.
- **Live smoke** (`test/live_smoke.sh`) — boots the built binary and confirms the full lifecycle over REST-TLS + MCP-stdio, capturing the transcript under `qa-results/<ts>/`.

Honest boundary: `brrotli-over-HTTP/3` is asserted programmatically in the Go integration test (a real h3 client); the `curl` live smoke uses the h1/h2 fallback of the same engine because `curl` does not speak HTTP/3 to a self-signed cert here.

6. Consumer wiring

- MCP registration: `constitution/mcp/scheduled-work-mcp.json`.

- Claude Code plugin: constitution/plugins/scheduled-work/ (.claude-plugin/plugin.json + .mcp.json + build.sh).
- Skill: constitution/skills/scheduled-work-queue/ (skill.md + register.sh).

7. Proposed governance-anchor handoff (for the conductor — NOT applied here)

Per the task's file-scope isolation, this component does NOT edit Constitution.md or any carrier. If the operator/conductor wishes to make this infrastructure a mandated capability, the following anchor text is PROPOSED for integration by the conductor (classification: universal, §11.4.17):

§11.4.NNN — Scheduled-work / reminder verify-before-report backing service. The REMINDER / BACKGROUND action-prefix (§11.4.140) MUST be backed by a durable scheduled-work queue (the scheduled-work-engine) so that deferred / background work is recorded with a lifecycle status, and a REMINDER MUST query list_needs_verification (blocked/uncertain/overdue) and re-confirm each item's real outcome with captured evidence (§11.4.5/§11.4.69) BEFORE the agent reports it done (§11.4.6/§11.4.108) — never marking an item done on assumption. The queue is decoupled + project-agnostic (§11.4.28), consumed by reference, and the loop done-condition (§11.4.87/§11.4.126) MUST NOT read satisfied while any tracked item is non-terminal. Recommended gate CM-SCHEDULED-WORK-VERIFY-BEFORE-DONE + paired §1.1 mutation. Composes §11.4.6/§11.4.28/§11.4.108/§11.4.116/§11.4.126/§11.4.140.

8. Cross-references

§11.4.6 · §11.4.8 · §11.4.28 · §11.4.30 · §11.4.77 · §11.4.81 · §11.4.108 · §11.4.116 · §11.4.126 · §11.4.140 · §11.4.164 · §1.1.